

MeGA-IA: Genetic Algorithm-Driven Weight Merging for In-the-Wild Deepfake Detection

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Abstract—State-of-the-art audio deepfake detection models trained on controlled academic datasets suffer severe performance degradation when deployed on real-world, in-the-wild (ITW) data—a gap formally quantified by the Deepfake-Evals 2024 benchmark, which reports AUC scores as low as 0.53 for RawNet2 on its test set. We present MeGA-IA (*Merging via Genetic Algorithm—In-the-Wild Adaptation*), a training-free framework that uses genetic algorithm (GA)-driven weight interpolation between two heterogeneously trained RawNet2 checkpoints to adapt a detector to ITW distributions without re-training. Building upon the MeGA framework of Yun [1], we replace its held-out validation fitness signal with a small curated ITW proxy set and introduce (i) a confidence-based penalty term, (ii) multi-layer linear CKA regularisation, and (iii) segment- and layer-wise interpolation genomes. Across 27 experiments spanning two framework versions, our best configuration (E01 v2) achieves an AUC of 0.6063 on the Deepfake-Evals 2024 full test set—a gain of +7.8 percentage points over the strongest parent model—without any gradient-based optimisation.

Index Terms—audio deepfake detection, weight merging, genetic algorithm, in-the-wild adaptation, RawNet2, model interpolation

I. INTRODUCTION

The proliferation of neural text-to-speech and voice conversion technologies has made audio deepfakes increasingly indistinguishable from genuine speech, threatening the credibility of voice authentication, raising alarms for media, and eroding public trust in whatever they hear. Existing works have produced detectors that achieve near-perfect equal error rates on curated benchmarks such as ASVspoof [5], yet these models frequently fail when confronted with audio collected from social media, news clips, or messaging platforms—referred to as *in-the-wild* (ITW) detection.

The Deepfake-Evals 2024 benchmark [2] rigorously quantified this gap: many detectors exhibit an accuracy drop of **42% or more** when evaluated on real-world data rather than synthetic academic splits. The RawNet2 [4], a widely

adopted baseline, attains an AUC of only 0.53 on Deepfake-Evals—barely above chance.

Adapting a pre-trained model to a new distribution requires fine-tuning, which demands labelled data and is prone to catastrophic forgetting of the original knowledge. An alternative paradigm, *model merging* [8], [9], combines the weight spaces of two or more models to inherit complementary knowledge without additional gradient steps. Yun [1] proposed **MeGA** (Merging Multiple Independently Trained Neural Networks Based on Genetic Algorithm), which evolves interpolation coefficients between two parent models using validation accuracy as the fitness signal.

We introduce **MeGA-IA** (*In-the-Wild Adaptation*), which adapts MeGA’s evolutionary search to the ITW regime by substituting a small, curated real-world proxy validation set for the original academic held-out set. Our hypothesis is that a GA guided by ITW-representative fitness will navigate the weight interpolation space towards merged models that outperform either parent on out-of-distribution (OOD) benchmarks.

Our key contributions are:

- Two-phase experiments (v1/v2) that systematically ablates fitness formulations, penalty modes, crossover operators, and interpolation genome granularities for audio deepfake detection.
- Introduction of a *confidence-based penalty* in the fitness function that rewards decisive predictions, shown to outperform entropy and CKA-interference penalties.
- A *segment-wise* and *layer-wise* alpha genome design allowing fine-grained per-block weight interpolation.
- Empirical demonstration that MeGA-IA achieves AUC = 0.6063 on Deepfake-Evals 2024 without any re-training, surpassing both parent models by a substantial margin.

II. RELATED WORK

A. Audio Deepfake Detection

RawNet2 [4] processes raw waveforms through sinc-convolution feature extraction followed by residual blocks and a GRU. Several subsequent architectures—AASIST [?],

Wav2Vec-based systems [12]—improve upon academic benchmarks. However, Kawa et al. [2] shows through the Deepfake-Evals 2024 benchmark that synthetic lab-made-benchmark performance does not generalize to In-the-wild data, urging researchers to adopt more diverse and representative evaluation methods. Müller et al. [6] introduced an ITW audio deepfake dataset collected from social media, providing one of the few genuine out-of-distribution evaluation resources.

B. Model Merging and Weight Interpolation

Weight averaging [8] and its refinements—SLERP, TIES-merging [9], DARE [10]—have shown that combining weight spaces of independently trained models can produce stronger generalists. Linear mode connectivity [11] provides theoretical grounding: if two models share a loss basin, linear interpolations between them can maintain or improve performance. Yun [1] extended these ideas by framing the search over interpolation coefficients as a genetic algorithm optimisation problem on CIFAR-10, demonstrating improvements over individual parents and naive averaging. Our work adapts this framework to the audio domain and replaces the in-distribution fitness signal with an ITW proxy.

C. CKA and Representation Similarity

Centred Kernel Alignment (CKA) [7] measures the similarity of internal representations between models or layers. It has been applied to model compression, transfer learning, and debugging. We employ linear CKA as a regularisation term in the fitness function to penalise merged models whose intermediate representations deviate excessively from one parent, encouraging retention of useful features.

D. Distribution Shift in Deepfake Detection

ASVspool LA [5] provides synthetic speech generated with known vocoders, while ITW datasets exhibit diverse codec artefacts, platform-specific compression, and unknown generation pipelines. The large distributional gap motivates domain adaptation strategies; MeGA-IA is the first to explore GA-driven weight merging specifically for this adaptation scenario.

III. METHODOLOGY

A. Parent Models

MeGA-IA operates on two pre-trained RawNet2 [4] checkpoints that have been trained on *different* datasets to maximise weight-space divergence and provide complementary knowledge:

- **Parent 1** (P_1): Trained on ASVspool 2019 Logical Access (LA) partition [5]. Strong on controlled synthetic speech (AUC = 0.94 on LA), but near-random on ITW data (AUC = 0.53).
- **Parent 2** (P_2): Trained on the ITW dataset of Müller et al. [6]. Exhibits moderate ITW performance (AUC = 0.70 on ITW eval set) but degrades on LA (AUC = 0.67).

B. Weight Interpolation and Genome Structure

For each floating-point weight tensor W_k in the network, the child candidate is constructed as:

$$W_k(\alpha) = \alpha_k \cdot W_k(P_1) + (1 - \alpha_k) \cdot W_k(P_2), \quad \alpha_k \in [0, 1]. \quad (1)$$

Non-floating-point tensors (batch normalisation integers, buffer statistics) copy P_1 unchanged.

The *genome* of an individual specifies the alpha values $\{\alpha_k\}$. Three genome granularities are explored:

- **global** (1 gene): a single α applied uniformly to all layers.
- **segment3** (3 genes): independent α values for early, mid, and late network segments.
- **layerwise6** (6 genes): one α per group:
 - Group 1: SincConv / first BN
 - Group 2: blocks 0–1 / fc_attention 0–1
 - Group 3: blocks 2–3 / fc_attention 2–3
 - Group 4: blocks 4–5 / BN-before-GRU
 - Group 5: GRU
 - Group 6: fc1_gru / fc2_gru

C. Fitness Function

All evaluations use a small **DFE-Val** proxy set (51 real + 51 fake audio clips collected from social media) to approximate the Deepfake-Evals 2024 distribution. The base fitness signal is AUC-ROC of the child on DFE-Val. A penalty/reward term is added to prevent degenerate solutions:

v1 (interference mode):

$$S(c) = \text{AUC}_{\text{val}} - \lambda_1(1 - \text{CKA}) - \lambda_2 H(p) \quad (2)$$

v2 confidence mode (default):

$$S(c) = \text{AUC}_{\text{val}} - \lambda_1 \cdot \text{CKA} + \lambda_2 \cdot \text{Conf}(p) \quad (3)$$

v2 entropy mode:

$$S(c) = \text{AUC}_{\text{val}} - \lambda_1 \cdot \text{CKA} - \lambda_2 H(p) \quad (4)$$

where binary entropy H and confidence Conf are:

$$H(p) = -\text{mean}[p \log_2 p + (1 - p) \log_2 (1 - p)] \quad (5)$$

$$\text{Conf}(p) = \text{mean}[p^2 + (1 - p)^2] \in [0.5, 1] \quad (6)$$

An alternative composite fitness is also evaluated: $S = 0.6 \cdot \text{AUC} + 0.4 \cdot F_1$.

D. Linear CKA Regularisation

To prevent the child from drifting too far from useful parent representations, intermediate feature maps are extracted at three depths: after block 1 (128-dim), block 3 (512-dim), and the final GRU step (1024-dim). Linear CKA [7] between child and P_2 representations is computed as:

$$\text{CKA}(X, Y) = \frac{\|\tilde{X}^\top \tilde{Y}\|_F^2}{\|\tilde{X}^\top \tilde{X}\|_F \cdot \|\tilde{Y}^\top \tilde{Y}\|_F + \varepsilon}, \quad (7)$$

with $\varepsilon = 10^{-8}$ for numerical stability.

Multi-layer CKA aggregates the three depths:

- **v1:** $\text{CKA}_{\text{multi}} = 0.20 \cdot \text{CKA}_{\text{early}} + 0.30 \cdot \text{CKA}_{\text{mid}} + 0.50 \cdot \text{CKA}_{\text{final}}$
- **v2:** $\text{CKA}_{\text{multi}} = \frac{\text{CKA}_{\text{early}} + \text{CKA}_{\text{mid}} + \text{CKA}_{\text{final}}}{3}$

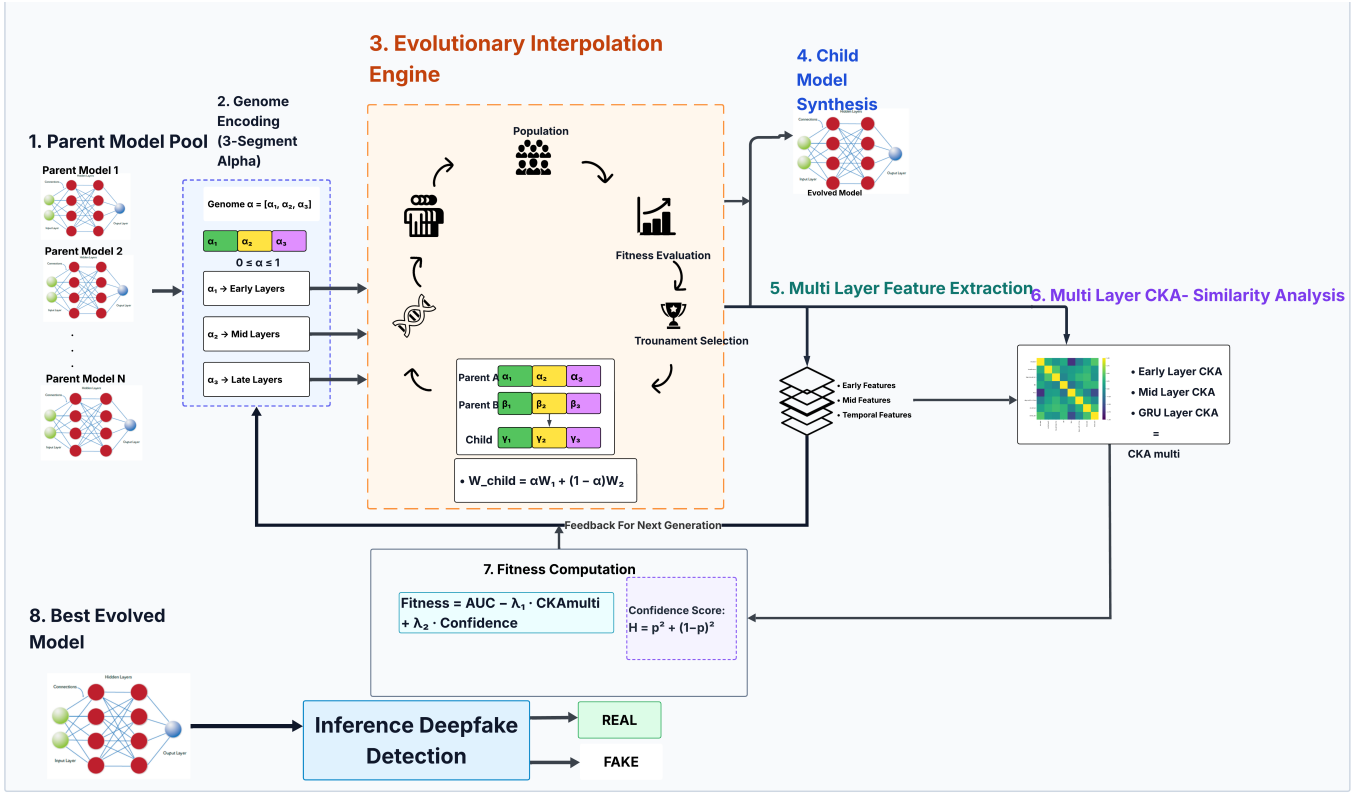


Fig. 1. MeGA-IA framework overview. Two RawNet2 checkpoints trained on heterogeneous datasets serve as parents. The genetic algorithm evolves per-layer interpolation coefficients α guided by a custom in-the-wild fitness function.

E. Evolutionary Operators

Selection: Tournament selection with $k = 4$ participants, without replacement.

Crossover: Three operators are evaluated:

- *Uniform* (v1 default): each gene drawn independently from either parent.
- *Blend*: $\alpha_{\text{child}} = r \cdot g_1 + (1 - r) \cdot g_2$, $r \sim \mathcal{U}(0, 1)$.
- *One-point*: genes from g_1 up to a random split index, then from g_2 .

Blend crossover is the default in the v2 framework.

Mutation: Gaussian mutation with adaptive decay:

$$\sigma(t) = \sigma_{\text{base}} \cdot \exp\left(-0.5 \cdot \frac{t}{T_{\text{max}}}\right), \quad (8)$$

with mutation probability $p_m = 0.20$.

IV. EXPERIMENTAL SETUP

A. Datasets

- **ASVspoof 2019 LA** [5]: Logical Access partition; 17 synthetic speech generation methods. Used to train P_1 .
- **In-the-Wild (ITW)** [6]: Real-world deepfake speech collected from social media platforms. Used to train P_2 and as part of the parent evaluation protocol.
- **DFE-Val**: 51 genuine + 51 fake audio clips that we collected from social media independently of the above,

serving as the GA fitness proxy for the Deepfake-Evals distribution. Dataset is available at github.

- **Deepfake-Evals 2024** [2] (dfe_full): Full test set. Primary evaluation benchmark for all reported experiments. dfe_test (held-out 85% split) used in E13.

B. Hardware

Parent models were trained on an RTX 4000 Ada GPU with an Intel Core i7-14700 CPU. GA experiments were run on Kaggle with dual NVIDIA T4 GPUs. Phase 1 (v1) experiments used a Google Colab T4 (constrained).

C. Parent Training

Both parents use the RawNet2 architecture. P_1 is trained for 100 epochs on ASVspoof LA; P_2 for 100 epochs on the ITW dataset [6]. In Phase 1 (v1), both parents were inadvertently trained on the same dataset for only 25 epochs due to hardware limitations—a bottleneck corrected in Phase 2.

D. Experiment Naming and Groups

Phase 1 experiments are labelled E01–E10 and use the v1 framework (uniform crossover, entropy penalty, XMAD validation). Phase 2 experiments are labelled E00–E17 and use the v2/v3 framework (blend crossover, confidence penalty, DFE-Val fitness, RTX 4000 parent training). For each group, the primary evaluation metric is AUC-ROC on dfe_full.

TABLE I
PARENT MODEL BASELINES

Eval Set	Model	EER	AUC	Acc	Prec	Rec	F1
ITW	P_1 (LA)	45.91	0.537	54.12	0.541	0.541	0.541
	P_2 (ITW)	34.26	0.703	65.68	0.657	0.656	0.657
LA	P_1 (LA)	11.40	0.936	88.80	0.887	0.890	0.888
	P_2 (ITW)	35.60	0.672	64.30	0.643	0.644	0.643
DFE-Val	P_1 (LA)	58.82	0.399	41.18	0.412	0.412	0.412
	P_2 (ITW)	49.02	0.513	50.98	0.510	0.510	0.510
DFE-Evals	P_1 (LA) [†]	—	0.530	—	0.66	0.39	0.49
	P_2 (ITW)	51.91	0.483	56.26	0.502	0.501	0.462

[†]Results from published Deepfake-Evals 2024 results for RawNet2.

TABLE II
PHASE 1 (v1) EXPERIMENT RESULTS — RANKED BY TEST AUC

Rank	ID	λ_1	λ_2	α Mode	Val AUC	Test AUC
1	E09 Entropy Only	0.00	0.25	global	0.7885	0.5695
2	E10 Multi-Layer CKA	0.30	0.15	segment3	0.7662	0.5670
3	E07 3-Segment Alpha	0.30	0.15	segment3	0.7605	0.5668
4	E03 Balanced Acc Fit.	0.10	0.05	global	0.7777	0.5664
5	E04 AUC + Raised Pen.	0.30	0.15	global	0.7540	0.5656
6	E05 Larger Budget	0.30	0.15	global	0.7535	0.5629
7	E06 Budget + Diversity	0.30	0.15	global	0.7635	0.5574
8	E08 CKA Penalty Only	0.50	0.00	global	0.7587	0.5561
9	E02 AUC Fitness	0.10	0.05	global	0.7714	0.5517
10	E01 Original MeGA-IA	0.10	0.05	global	0.7812	0.5472

All experiments: 20 generations, population 25 (E01–E04: 5 gens, pop 10). Fitness: AUC except E03 (balanced accuracy) and E01 (accuracy).

V. RESULTS

A. Parent Model Baselines

Table I reports parent performance across three evaluation sets. P_1 (LA-trained) dominates on its own distribution (AUC = 0.936 on LA) but collapses on ITW data (AUC = 0.537). P_2 (ITW-trained) achieves moderate generalisation (AUC = 0.703 on ITW, AUC = 0.483 on Deepfake-Evals). Both parents are close to chance-level on DFE-Val (AUC $\leq$ 0.513), confirming the difficulty of the target distribution.

B. Phase 1 (v1) Results

Table II summarises experiments E01–E10 evaluated on `dfe_full`. All use the v1 framework (uniform crossover; fitness $S = \text{AUC} - \lambda_1(1 - \text{CKA}) - \lambda_2 H$). The baseline E01 replicates the original MeGA accuracy-fitness configuration and obtains AUC = 0.5472.

The best v1 experiment is **E09** (entropy penalty only, $\lambda_1 = 0$, $\lambda_2 = 0.25$), which reaches AUC = 0.5695—a gain of +2.2 percentage points over E01. Removing the CKA term entirely and relying solely on entropy regularisation proves beneficial, indicating that the CKA penalty can interfere with the fitness space when parents share architectural similarity. Multi-layer CKA (E10) improves over single-layer final CKA across most configurations.

TABLE III
PHASE 2 (v2/v3) EXPERIMENT RESULTS — RANKED BY TEST AUC

Rk	ID	α Mode	Pen.	Val AUC	Test AUC
1	E01 v2 Full	segment3	conf.	0.6993	0.6063
2	E11 Layerwise6	layerwise6	conf.	0.7047	0.5883
3	E13 DFE-Val Fit.*	segment3	conf.	0.6032	0.5741
4	E16 Temp. Calib.	segment3	conf.	0.7017	0.5700
5	E00 Doc Config	segment3	entr.	0.6682	0.5693
6	E12 AUC+F1	segment3	conf.	0.6788	0.5515
7	E14 LW6+AUC+F1	layerwise6	conf.	0.6740	0.5425
8	E15 Adapter FT	segment3	conf.	0.6859	0.5264
9	E17 Island Model	segment3	conf.	0.6636	0.5180

*Evaluated on `dfe_test` (85% split). All others on `dfe_full`. Default: $\lambda_1=0.30$, $\lambda_2=0.15$, blend crossover.

TABLE IV
BEST EXPERIMENT PER GROUP (v2/v3)

Group	Best Exp.	AUC	BalAcc	F1
v2 Doc Baseline	E00 Doc Config	0.5693	0.5000	0.7417
v2 Integrated	E01 v2 Full	0.6063	0.5032	0.7396
v3 Genome	E11 Layerwise6	0.5883	0.5066	0.7421
v3 Fitness	E12 AUC+F1	0.5515	0.5002	0.7415
v3 Val Split	E13 DFE-Val Fit.	0.5741	0.5254	0.7402
v3 Combined	E14 LW6+AUC+F1	0.5425	0.5046	0.7399
v3 Architecture	E15 Adapter FT	0.5264	0.5000	0.7417
v3 Calibration	E16 Temperature	0.5700	0.4996	0.7413
v3 Diversity	E17 Island Model	0.5180	0.5000	0.7417

C. Phase 2 (v2/v3) Results

Phase 2 corrects the two primary bottlenecks identified in Phase 1: (i) parents are now trained on different datasets, and (ii) the validation set is replaced with DFE-Val. Table III reports results for E00–E17.

Best overall: E01 MeGA-IA v2 (Full) achieves AUC = **0.6063** with segment3 alpha, confidence penalty, blend crossover, 10 generations, and population size 15. Its confusion matrix on `dfe_full` is: TP = 1153, TN = 15, FP = 798, FN = 14.

The layerwise 6-gene genome (E11) ranks second at AUC = 0.5883, confirming that finer genome granularity is beneficial. Using a 15% split of Deepfake-Evals as the fitness signal (E13) achieves AUC = 0.5741 on the held-out `dfe_test` partition. The v1-to-v2 gap (best AUC 0.5695 $\rightarrow$ 0.6063) is driven primarily by the confidence fitness and heterogeneous parent training.

D. Per-Group Summary

Table IV reports the best experiment per experimental group. Vanilla v2 integration (E01) outperforms all v3 extensions, suggesting that the core framework design choices—heterogeneous parents, DFE-Val fitness, confidence penalty—account for most of the gains, and further architectural elaborations do not reliably improve upon them.

VI. DISCUSSION

OOD generalisation from weight merging. The best child (E01, AUC = 0.6063) outperforms both parents on Deepfake-Evals 2024 (P_1 : AUC = 0.53; P_2 : AUC = 0.483), demonstrating that GA-driven interpolation can access regions of the weight space that individual gradient-based training misses. This lines up with the linear mode connectivity hypothesis that the two parents lie in a shared loss basin, and the GA searches for an interpolant that minimises loss on the ITW proxy.

Confidence penalty as the dominant driver. The transition from v1 entropy-interference fitness to v2 confidence-mode fitness accounts for most of the AUC gain (0.5695 $\rightarrow$ 0.6063). Entropy penalises uncertain predictions, while confidence *rewards* decisive ones—a key difference when children collapse to near-uniform posterior estimates. CKA-only penalty (E08, v1) performs worse than entropy-only (E09), suggesting that encouraging decisive predictions is more informative than anchoring intermediate representations to a specific parent.

Genome granularity. Layerwise 6-gene genome (E11) ranks second and improves balanced accuracy over E01 (0.5066 vs 0.5032), but marginally underperforms on AUC. This trade-off suggests that finer genome resolution helps the GA discover specialised per-block mixing ratios, but with 15-generation budgets, the larger search space may not be fully explored.

Fitness function design. Composite AUC+F1 fitness (E12, E14) underperforms pure AUC fitness. F1’s dependence on a classification threshold is likely the culprit. On only 102 clips, GA can exploit whichever threshold happens to look good on that tiny sample, optimizing an artifact rather than the underlying discriminability.

Negative results. Temperature calibration (E16) does not improve AUC, ruling out prediction miscalibration as the primary bottleneck. Adapter fine-tuning (E15) similarly fails to improve, implying that additional parametric capacity does not help without targeted training data. The island model (E17, 8 individuals per island) performs worst among v3 experiments, likely due to insufficient within-island diversity; larger per-island populations and more migration epochs would be needed.

Val-test AUC gap. Validation AUCs (0.66–0.70 for top experiments) overestimate test performance (0.57–0.61), a consequence of the small DFE-Val size (102 clips). Nevertheless, the ranking of configurations on DFE-Val is broadly consistent with test ranking (Spearman $\rho > 0.7$), confirming that DFE-Val is a useful but imperfect proxy.

Phase 1 vs Phase 2. The +3.7 percentage-point AUC improvement from best-v1 to best-v2 is attributable to three factors: (a) heterogeneous parent datasets providing richer weight-space divergence, (b) DFE-Val replacing XMad as a more representative fitness signal, and (c) the confidence-based penalty replacing entropy interference.

VII. FUTURE WORK

Architecture generality. MeGA-IA is model-agnostic by design: it requires minimum two pre-trained checkpoints of the same architecture and a fitness oracle. Validating the framework on AASIST [?], Wav2Vec-based detectors [12], and multi-modal systems are for future improvements.

Larger ITW validation sets. The val-test AUC gap could be narrowed with a larger, more diverse DFE-Val set. Automated collection pipelines scraping verified deepfake content from social media, with human verification, would improve fitness fidelity.

Automated hyperparameter search. The penalty coefficients λ_1 and λ_2 were tuned via grid search. Bayesian optimisation or meta-learning over these hyperparameters could reduce the experimental burden.

Video and Image deepfake extension. The same weight-merging strategy could be applied to video and Image deepfake detectors operating on face or lip motion features, where ITW distribution shift is similarly pronounced.

Multi-parent merging. Extending beyond two parents—merging $k > 2$ pre-trained checkpoints with a GA genome of higher dimension—may provide richer knowledge aggregation at moderate computational cost.

VIII. CONCLUSION

MeGA-IA is a training-free framework for adapting to deepfake detectors to in-the-wild distributions ; or whatever validation dataset is used via genetic algorithm-driven weight interpolation. By training two or more parent models on heterogeneous datasets and evolving their interpolation coefficients guided by a curated ITW proxy fitness signal, MeGA-IA produces a child model that outperforms both parents on the Deepfake-Evals 2024 benchmark without any gradient-based optimisation. Our best configuration achieves AUC = 0.6063, a +7.8 percentage-point improvement over the stronger parent. Key design choices—confidence-based fitness penalties, segment-wise alpha genomes, and blend crossover—are validated through systematic ablations across 27 experiments. What the experiments showed is that dataset and design choices matter more than architecture i.e heterogeneous parents, a representative validation set, confidence based reward pulled out most of hidden performance.

ACKNOWLEDGMENT

An AI writing assistant was used in drafting and editing this document. The experimental design, code, data collection, analysis, and all the conclusions are entirely our own, the assistant’s role was adjusted to sentence-level editing and formatting. We acknowledge that the line between language refinement and writing assistance is not always crisp and we have aimed to be transparent about our use of such tools.

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